

The logic of arguments - Set 3 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	A	The logic of arguments

Worked solutions

Question 1 LGC-15 · The logic of arguments · spec Arg2

Let P : 'the integer n is divisible by 4' and Q : 'the integer n^2 is divisible by 8'. Which one of the following is true?

Answer: A

If $4 \mid n$ then $16 \mid n^2$, so $8 \mid n^2$: P is sufficient. Conversely, if $8 \mid n^2$ then n is even, say $n = 2m$, and $8 \mid 4m^2$ forces m to be even, so $4 \mid n$: P is also necessary. (Tempting wrong answer: 'sufficient but not necessary' - but no counterexample exists.)